Example: Applying PCA to a Simple 2D Dataset

Consider a dataset with two features:

$$X = \begin{bmatrix} 2 & 3 \\ 3 & 4 \\ 4 & 5 \\ 5 & 6 \\ 6 & 7 \end{bmatrix}$$

We will perform **Principal Component Analysis (PCA)** to reduce the dataset from **2D to 1D**.

Step 1: Compute the Mean of Each Feature

For a dataset X with n samples and d features:

$$\mu_j = \frac{1}{n} \sum_{i=1}^{n} X_{ij}$$

For our dataset:

$$\mu_1 = \frac{2+3+4+5+6}{5} = 4, \quad \mu_2 = \frac{3+4+5+6+7}{5} = 5$$

Mean vector:

$$\mu = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Step 2: Center the Data (Subtract the Mean)

Each data point is transformed as:

$$X_{\text{centered}} = X - \mu$$

$$X_{\text{centered}} = \begin{bmatrix} 2-4 & 3-5 \\ 3-4 & 4-5 \\ 4-4 & 5-5 \\ 5-4 & 6-5 \\ 6-4 & 7-5 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ -1 & -1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix}$$

Step 3: Compute the Covariance Matrix

Covariance matrix formula:

$$\Sigma = \frac{1}{n-1} X_{\text{centered}}^T X_{\text{centered}}$$

For our dataset:

$$\sigma_{11} = \frac{1}{4} \sum_{i=1}^{5} x_{i1}^2 = \frac{1}{4}(4+1+0+1+4) = 2.5$$

$$\sigma_{22} = \frac{1}{4}\sum_{i=1}^{5} x_{i2}^2 = 2.5$$

$$\sigma_{12} = \frac{1}{4}\sum_{i=1}^{5} x_{i1}x_{i2} = \frac{1}{4}(4+1+0+1+4) = 2.5$$

$$\Sigma = \begin{bmatrix} 2.5 & 2.5 \\ 2.5 & 2.5 \end{bmatrix}$$

Step 4: Compute Eigenvalues and Eigenvectors

The characteristic equation is:

$$\det(\Sigma - \lambda I) = 0$$

$$\begin{vmatrix} 2.5-\lambda & 2.5 \\ 2.5 & 2.5-\lambda \end{vmatrix} = (2.5-\lambda)(2.5-\lambda) - (2.5 \times 2.5) = 0$$

$$\lambda^2 - 5\lambda = 0$$

$$\lambda(\lambda - 5) = 0$$

Thus, the eigenvalues are:

$$\lambda_1 = 5, \quad \lambda_2 = 0$$

Eigenvectors are found by solving $(\Sigma - \lambda I)v = 0$:

For $\lambda_1 = 5$:

$$\begin{bmatrix} 2.5-5 & 2.5 \\ 2.5 & 2.5-5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving:

$$-2.5v_1 + 2.5v_2 = 0 \Rightarrow v_1 = v_2$$

So, the eigenvector is:

$$v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 0$:

$$v_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Thus, the eigenvectors are:

$$P = \begin{bmatrix} 0.7071 & -0.7071 \\ 0.7071 & 0.7071 \end{bmatrix}$$

Step 5: Project Data onto the First Principal Component

The transformed data is:

$$X_{\text{projected}} = X_{\text{centered}} \cdot v_1$$

$$= \begin{bmatrix} -2 & -2 \\ -1 & -1 \\ 0 & 0 \\ 1 & 1 \\ 2 & 2 \end{bmatrix} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} -2-2 \\ -1-1 \\ 0+0 \\ 1+1 \\ 2+2 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -4 \\ -2 \\ 0 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} -2.83 \\ -1.41 \\ 0 \\ 1.41 \\ 2.83 \end{bmatrix}$$

Final Output: Reduced 1D Representation

The original 2D dataset is now represented in **1D**:

$$\begin{bmatrix} -2.83 \\ -1.41 \\ 0 \\ 1.41 \\ 2.83 \end{bmatrix}$$

The PCA process successfully reduces the dimensionality from **2D to 1D**, preserving the maximum variance in the data.